

DD: OSR-RMA-01-FLOW-SWAP-GPON — GPON ONT swap (remote-first, field fallback)

Developer implementation guide: DD_OSR-RMA-01-FLOW-SWAP-GPON-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the operator BPMN flow DD_OSR-RMA-01-FLOW-SWAP-GPON-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** GPON Equipment Swap Flow
Satellite of: OSR-RMA-01 Equipment Swap & RMA Per-flow satellite for the GPON ONT swap (swap_kind=GPON_REMOTE or GPON_FIELD). Owns the BPMN EquipmentSwap_GPON_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups. v1.0 scope: KE WIK only. Remote-first happy path with field fallback. WUG/WTZ/YASSN variants via separate BPMN files when workshops land.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/08_osr_equipment/DD_OSR-RMA-01-FLOW-SWAP-GPON-v1.0.md
Version	v1.0
DD type	operator BPMN flow
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- PATCH /api/homepasses/hp_01HZ_NRB_KILELESHWA_42B/network-path/ports
- PATCH /api/homepasses/{id}/network-path/ports
- POST /api/equipment-instances/inst_01HZ_ONT_WANJIKU_NEW/lifecycle-events
- POST /api/equipment-instances/inst_01HZ_ONT_WANJIKU_OLD/lifecycle-events
- POST /api/notifications

- POST /api/swap-requests/gpon
- POST /nms-adapter/v1/ont/bind
- POST /nms-adapter/v1/ont/unassign
- POST /webhook/nms/ont-online

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- contractor-stock-route
- customer-ont-online
- homepass-network-path-update
- instance-state-transition
- nms-remote-bind-stub
- nms-remote-unassign-stub
- noc-confirm-remote-success
- notification-send
- nrb-olt-01
- nrb-olt-02
- swap-dispatchers
- wik-internal-tech-team

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.

4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** GPON Equipment Swap Flow

Satellite of: OSR-RMA-01 Equipment Swap & RMA

Per-flow satellite for the GPON ONT swap (swap_kind=GPON_REMOTE or GPON_FIELD). Owns the BPMN EquipmentSwap_GPON_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups.

v1.0 scope: KE WIK only. Remote-first happy path with field fallback. WUG/WTZ/YASSN variants via separate BPMN files when workshops land.

1. What this flow is

When a customer's GPON ONT fails — or NOC detects an imminent failure, or the customer requests an in-tech upgrade — there are two paths:

- **Remote-first (preferred):** NOC unassigns the old ONT serial in the NMS (Network Management System for the OLT), couriers / customer-picks-up a new ONT, customer self-installs the new ONT in their existing fiber socket, NMS auto-discovers the new ONT serial and binds it. No truck roll. Recovery of the defective unit happens at next courier exchange or shop visit.
- **Field fallback:** if customer can't or won't self-swap (elderly customer, customer doesn't open door for courier, customer reports doesn't understand the installation steps, etc.), the BPMN advances to field-visit pattern (same as SWAP-HFC but with NMS calls instead of Broadhub).

Either path drives a HomePass update: the HomePass's network_path terminal node carries the ONT serial as its code; after the swap, that code must be updated to the new ONT serial. The homepass-network-path-update worker handles this at TARGET_PLACED.

A GPON swap creates two swap_kind values: GPON_REMOTE is created when the BPMN starts (default for GPON), and the swap-request is updated in-place to GPON_FIELD if/when the BPMN crosses the field-fallback boundary. Both kinds share the same equipment_swap_request row; only the swap_kind discriminator and flow_payload.remoteAttemptCount differ.

What this flow does NOT do

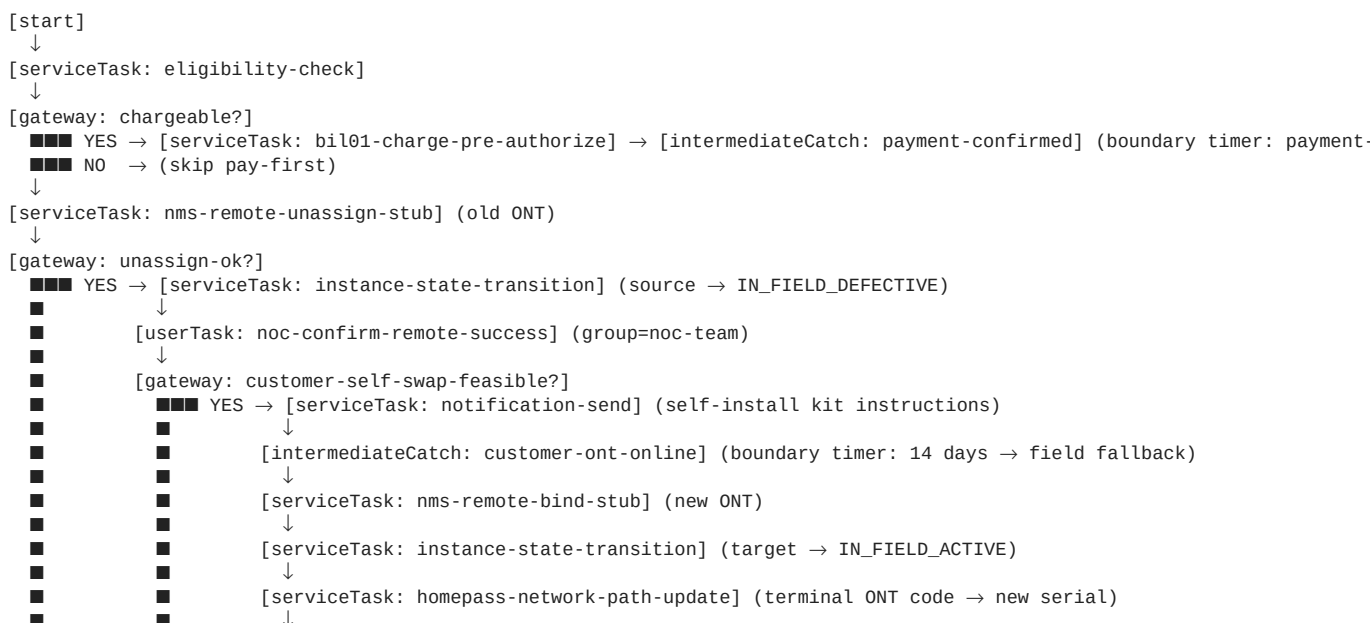
- **Does not handle HFC equipment swaps.** Use swap_kind=HFC_FIELD (separate FLOW-SWAP-HFC satellite).
- **Does not handle equipment termination retrieval** — that's EQP_TERMINATION (separate FLOW-EQP satellite).
- **Does not validate slot/port preservation.** GPON doesn't bind to a chassis slot/port for authorization (the OLT auto-discovers the serial). The OLT F/S/P (Frame/Slot/Port) is recorded as instance metadata but is not preservation-required; if the customer relocates within the same building and the new ONT lands on a different OLT port, the swap still completes (the OLT auto-discovery handles it). Relocation across HomePasses is a different flow (subscription relocation, not swap).
- **Does not handle the courier / shop pickup logistics.** That's a separate workflow (TBD pending workshops on Wananchi's self-install kit fulfillment chain). This flow assumes the new ONT lands in the customer's hands via some out-of-band channel and the customer self-installs.

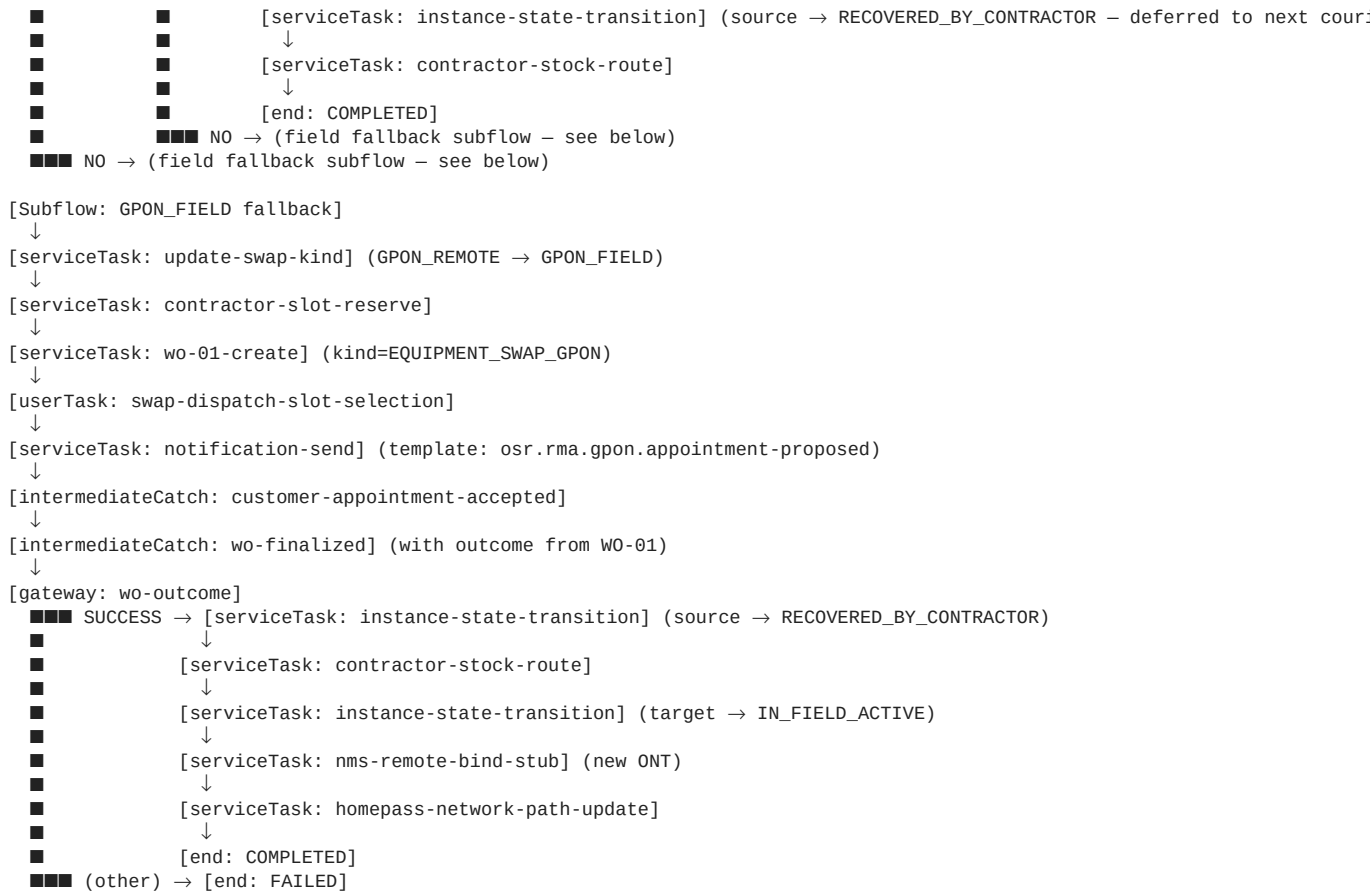
Glossary additions (specific to this flow)

Term	Meaning
OLT F/S/P	OLT Frame / Slot / Port — the physical port on the GPON Optical Line Terminal to which the customer's fiber drop is connected. Used as instance metadata for traceability; not used for authorization.
NMS remote unassign	The NMS API call that removes an ONT serial's authorization from the OLT. The fiber link remains; the ONT just stops being recognized.
NMS auto-discovery	When a new ONT is powered on at a customer's premises and the OLT detects its serial, NMS auto-binds it if the serial is pre-registered. Wananchi runs the malogs portal (malogs.wananchi.com/gpon_auto_find/) which lists discovered ONTs with their F/S/P.
Self-install kit	A pre-packaged box with new ONT + cables + 1-page instructions, sent to the customer via courier (or picked up at a Wananchi shop). Customer plugs in the new ONT; the OLT auto-discovers + NMS auto-binds (if the serial is pre-registered for the customer's account).
Customer self-swap	Customer physically replaces the ONT themselves following the self-install kit instructions. Does not require a Wananchi technician on site.

2. BPMN — EquipmentSwap_GPON_WIK.bpmn

2.1 Process shape





2.2 BPMN XML — key snippets

Remote unassign with retry logic (up to 3 attempts):

```

<bpmn:serviceTask id="Task_NmsRemoteUnassign" name="NMS remote unassign source ONT"
  camunda:type="external" camunda:topic="nms-remote-unassign-stub">
  <bpmn:extensionElements>
    <camunda:inputOutput>
      <camunda:inputParameter name="instanceId">${sourceInstanceId}</camunda:inputParameter>
      <camunda:inputParameter name="ontSerial">${flowPayload.sourceOntSerial}</camunda:inputParameter>
      <camunda:inputParameter name="oltFrameSlotPort">${flowPayload.oltFsp}</camunda:inputParameter>
    </camunda:inputOutput>
  </bpmn:extensionElements>
</bpmn:serviceTask>

<bpmn:boundaryEvent id="Timer_RemoteAttemptTimeout" attachedToRef="Task_NmsRemoteUnassign" cancelActivity="false">
  <bpmn:timerEventDefinition>
    <bpmn:timeDuration>PT2M</bpmn:timeDuration>
  </bpmn:timerEventDefinition>
</bpmn:boundaryEvent>

```

HomePass update:

```

<bpmn:serviceTask id="Task_HomepassUpdate" name="Update HomePass terminal ONT node"
  camunda:type="external" camunda:topic="homepass-network-path-update">
  <bpmn:extensionElements>
    <camunda:inputOutput>
      <camunda:inputParameter name="accountId">${accountId}</camunda:inputParameter>
      <camunda:inputParameter name="sourceInstanceId">${sourceInstanceId}</camunda:inputParameter>
      <camunda:inputParameter name="targetInstanceId">${targetInstanceId}</camunda:inputParameter>
      <camunda:inputParameter name="swapKind">${swapKind}</camunda:inputParameter>
    </camunda:inputOutput>
  </bpmn:extensionElements>
</bpmn:serviceTask>

```

Field-fallback timer (14 days waiting for customer self-install):

```

<bpmn:boundaryEvent id="Timer_SelfSwapTimeout" attachedToRef="Catch_CustomerOntOnline" cancelActivity="true">
  <bpmn:timerEventDefinition>
    <bpmn:timeDuration>P14D</bpmn:timeDuration>
  </bpmn:timerEventDefinition>
</bpmn:boundaryEvent>

```

2.3 Per-operator BPMN composition

KE WIK ships EquipmentSwap_GPON_WIK.bpmn. WTZ is fiber-only (GPON) per workshop sources — likely needs its own variant with longer self-install timeout (rural connectivity) when workshops land. YASSN: Senegal GPON variant TBD pending workshop on self-install kit availability (Senegal may not have shop pickup as widely as Nairobi).

3. Flow_payload shape (GPON_REMOTE / GPON_FIELD)

```
{
  "olt": "nrb-olt-01",
  "oltFsp": {"frame": 1, "slot": 2, "port": 3},
  "sourceOntSerial": "48575443ABCD1234",
  "targetOntSerial": "48575443BEEF5678",
  "sourceVendor": "Huawei",
  "targetVendor": "Huawei",
  "remoteAttemptCount": 0,
  "maxRemoteAttempts": 3,
  "selfInstallKitDispatched": false,
  "selfInstallKitTrackingRef": null,
  "fieldFallbackTriggered": false,
  "fieldFallbackReason": null
}
```

Field meanings:

Field	Sourced from	Purpose
olt, oltFsp	Source instance network_slot_metadata	OLT identity for NMS calls; F/S/P recorded for traceability (not preservation-required for GPON)
sourceOntSerial, targetOntSerial	Source/target instance serial_number	The identifier the NMS adapter unbinds / binds
sourceVendor, targetVendor	PLM-CFG-06 SKU manufacturer	NMS adapter may need vendor-specific call paths (Huawei vs FiberHome OLTs in WIK fleet)
remoteAttemptCount	Incremented by NMS worker on each retry	Drives field-fallback decision when reaches maxRemoteAttempts
maxRemoteAttempts	Operator config (default 3)	Field fallback trigger threshold
selfInstallKitDispatched, selfInstallKitTrackingRef	NOT-01 or external courier system	Audit trail for the kit logistics
fieldFallbackTriggered, fieldFallbackReason	Set when subflow enters GPON_FIELD path	Audit + reporting

4. Rules (GPON-specific)

Rule	Description
R-GPON-1	Source equipment_kind MUST be ONT. Eligibility worker rejects with failureCode=NOT_GPON_FAMILY otherwise.
R-GPON-2	Source network_slot_metadata.olt MUST be non-null. Eligibility worker rejects with failureCode=MISSING_OLT_REF otherwise — NMS adapter can't address the unbind without OLT identity.
R-GPON-3	NMS remote unassign retry: up to flow_payload.maxRemoteAttempts (default 3) attempts before transitioning to field fallback. Each retry increments remoteAttemptCount.
R-GPON-4	Self-install timeout: 14-day boundary timer on customer-ont-online message catch. If timer fires, BPMN transitions to field fallback (NOT to FAILED — Wananchi prefers to escalate to truck roll rather than abandon the customer).
R-GPON-5	HomePass terminal-node update (R-OSR-RMA-16 enforces): homepass-network-path-update worker MUST succeed at TARGET_PLACED. If it fails (HomePass not found, PATCH rejected by HomePass-CFG, etc.), swap-request → FAILED with transition_reason=HOMEPASS_NETWORK_PATH_UPDATE_FAILED. Source instance state rolled back to IN_FIELD_DEFECTIVE. NMS bind not rolled back (the customer's new ONT is online; manual reconciliation step needed to fix HomePass — Ops escalation).
R-GPON-6	If account is not bound to a HomePass (no account.homepass_id — rare, possibly migration data), homepass-network-path-update worker skips the call and writes networkPathUpdated=false with reason=NO_HOMEPASS_BINDING. Swap proceeds to COMPLETED. ICN-01 notifies swap-dispatchers of the unbound account for review.

Rule	Description
R-GPON-7	Cross-OLT swap (source on OLT nrb-olt-01, target serial pre-registered on OLT nrb-olt-02) is rejected at eligibility with failureCode=CROSS_OLT_NOT_SUPPORTED. v1.0 scope: same-OLT swaps only. Cross-OLT moves are subscription relocations, not swaps.
R-GPON-8	When the swap transitions to field fallback, the swap-request's swap_kind is updated in-place from GPON_REMOTE to GPON_FIELD. History row records the transition with transition_reason=REMOTE_TO_FIELD_FALLBACK and the trigger reason (max_attempts_reached / customer_self_swap_declined / self_install_timeout).

5. REST API

POST /api/swap-requests/gpon HTTP/1.1
Authorization: Bearer <noc-jwt>
Content-Type: application/json

```
{
  "operatorCode":      "WIK",
  "accountId":         "acct_01HZ_WANJIKU",
  "subscriptionId":    "sub_01HZ_WANJIKU_FIBER",
  "sourceInstanceId":  "inst_01HZ_ONT_WANJIKU_OLD",
  "targetInstanceId":  "inst_01HZ_ONT_WANJIKU_NEW",
  "reasonCode":        "CUSTOMER_REPORTED_DEFECT",
  "initiatedByUserId": "noc-agent-wik-002",
  "notes":             "Customer reports ONT power LED off, no internet. Tested with reboot – no recovery."
}

→ 201 Created
{
  "swapRequestId":      "swap_01HZ_WANJIKU_ONT",
  "operatorCode":       "WIK",
  "swapKind":           "GPON_REMOTE",
  "currentState":       "ELIGIBILITY_CHECK",
  "accountId":          "acct_01HZ_WANJIKU",
  "sourceInstanceId":   "inst_01HZ_ONT_WANJIKU_OLD",
  "targetInstanceId":   "inst_01HZ_ONT_WANJIKU_NEW",
  "reasonCode":         "CUSTOMER_REPORTED_DEFECT",
  "initiatedByUserId":  "noc-agent-wik-002",
  "initiatedAt":        "2026-05-22T09:15:00Z",
  "bpmnProcessInstanceId": "camunda-pi-8a4b1c9d"
}
```

6. Kafka events emitted

Topic prefix: sophix.osr.rma.*.

6.1 SwapRequestSourceRecovered (remote-first path: deferred to next courier exchange)

```
{
  "eventType":  "SwapRequestSourceRecovered",
  "aggregateId": "swap_01HZ_WANJIKU_ONT",
  "timestamp":  "2026-06-05T11:00:00.000Z",
  "payload": {
    "operatorCode":      "WIK",
    "swapRequestId":     "swap_01HZ_WANJIKU_ONT",
    "swapKind":          "GPON_REMOTE",
    "sourceInstanceId":  "inst_01HZ_ONT_WANJIKU_OLD",
    "sourceSerial":      "48575443ABCD1234",
    "recoveryContractorId": "wik-internal-tech-team",
    "recoveredAt":       "2026-06-05T11:00:00+03:00",
    "recoveryChannel":   "COURIER_PICKUP"
  }
}
```

6.2 SwapRequestHomePassUpdated

```
{
  "eventType":  "SwapRequestHomePassUpdated",
  "aggregateId": "swap_01HZ_WANJIKU_ONT",
  "timestamp":  "2026-05-25T10:30:15.000Z",
  "payload": {
```

```

    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_WANJIKU_ONT",
    "swapKind": "GPON_REMOTE",
    "accountId": "acct_01HZ_WANJIKU",
    "homepassId": "hp_01HZ_NRB_KILELESHWA_42B",
    "previousOntSerial": "48575443ABCD1234",
    "newOntSerial": "48575443BEEF5678",
    "networkPathUpdated": true
  }
}

```

6.3 SwapRequestCompleted

```

{
  "eventType": "SwapRequestCompleted",
  "aggregateId": "swap_01HZ_WANJIKU_ONT",
  "timestamp": "2026-05-25T10:35:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_WANJIKU_ONT",
    "swapKind": "GPON_REMOTE",
    "accountId": "acct_01HZ_WANJIKU",
    "subscriptionId": "sub_01HZ_WANJIKU_FIBER",
    "sourceInstanceId": "inst_01HZ_ONT_WANJIKU_OLD",
    "targetInstanceId": "inst_01HZ_ONT_WANJIKU_NEW",
    "recoveryContractorId": "wik-internal-tech-team",
    "woId": null,
    "chargeCode": null,
    "customerChargeAmount": null,
    "homepassId": "hp_01HZ_NRB_KILELESHWA_42B",
    "closedAt": "2026-05-25T10:35:00+03:00",
    "completionPath": "REMOTE_SUCCESS"
  }
}

```

7. Worked end-to-end example — Wanjiku's ONT swap (remote-first happy path)

Customer: Wanjiku, WIK fiber subscriber (GPON, single-play internet, in warranty — purchased 6 months ago). Account acct_01HZ_WANJIKU bound to HomePass hp_01HZ_NRB_KILELESHWA_42B. ONT inst_01HZ_ONT_WANJIKU_OLD serial 48575443ABCD1234, on OLT nrb-olt-01 F1/S2/P3. Recovery contractor: wik-internal-tech-team (internal team handles GPON remote swaps in Nairobi central). New ONT inst_01HZ_ONT_WANJIKU_NEW (serial 48575443BEEF5678) is in WIK main warehouse, pre-registered for Wanjiku's account.

Day 0 — 09:15: NOC receives Wanjiku's call (ONT power LED off). NOC agent posts POST /api/swap-requests/gpon. Response 201: state ELIGIBILITY_CHECK.

Day 0 — 09:15:30: Eligibility worker reads source instance metadata (OLT present), validates target SKU (Huawei EG8145V5 = same model), confirms warranty (purchase < 1 year, in warranty). Sets chargeable=false, chargeCode=null. Builds flow_payload. State advances directly to AWAITING_SLOT (no pay-first because not chargeable).

Wait — that's wrong for the remote path. Remote-first doesn't reserve a slot. Eligibility worker recognizes GPON_REMOTE and advances directly to REMOTE_UNASSIGN_IN_PROGRESS (skipping AWAITING_SLOT). State machine allows ELIGIBILITY_CHECK → REMOTE_UNASSIGN_IN_PROGRESS for GPON_REMOTE per framework.

Day 0 — 09:15:35: Worker nms-remote-unassign-stub fires:

```

POST /nms-adapter/v1/ont/unassign
{"ontSerial":"48575443ABCD1234","oltFsp":{"frame":1,"slot":2,"port":3},"olt":"nrb-olt-01"}
→ 200 {"unassignOk":true,"adapterResponseCode":"OK","unassignedAt":"2026-05-22T09:15:35Z"}

```

Worker instance-state-transition advances source instance:

```

POST /api/equipment-instances/inst_01HZ_ONT_WANJIKU_OLD/lifecycle-events
{"toState":"IN_FIELD_DEFECTIVE","toAccountId":"acct_01HZ_WANJIKU",
 "reasonCode":"REMOTE_UNASSIGN_OK","contractorId":"wik-internal-tech-team",
 "originRefKind":"swap_request","originRefId":"swap_01HZ_WANJIKU_ONT",
 "actorUserId":"noc-agent-wik-002","expectedEventSequence":5}
→ 201 {"eventId":"evt_01HZ_WANJIKU_DEFECT","eventSequence":6}

```

State REMOTE_UNASSIGN_IN_PROGRESS → SOURCE_RECOVERED. (Note: the physical unit is still at premises; "RECOVERED" in instance-lifecycle means logically unassigned from NMS, not physically recovered. Physical recovery happens later via courier exchange — handled in recoveryChannel=COURIER_PICKUP later.)

Actually — per state machine, the lifecycle event toState is IN_FIELD_DEFECTIVE not RECOVERED_BY_CONTRACTOR. Swap-request state is SOURCE_RECOVERED (workflow concept), but instance state stays at IN_FIELD_DEFECTIVE until physical pickup.

Day 0 — 09:16: NOC user task noc-confirm-remote-success created. NOC agent reviews: customer Wanjiku is tech-savvy (per CRM notes), shop pickup available 3km from her location, kit ready. NOC agent submits: customerSelfSwapConfirmed=true. State remains SOURCE_RECOVERED.

Day 0 — 09:20: Worker notification-send fires:

```
POST /api/notifications
{"recipientKind":"customer","recipientRef":"acct_01HZ_WANJIKU",
 "templateId":"osr.rma.gpon.self-install-kit-ready","templateVars":{"
  "shopAddress":"Wananchi Shop Westgate, 2nd Floor",
  "shopHours":"Mon-Sat 09:00-18:00","ontSerial":"48575443BEEF5678",
  "instructionsUrl":"https://help.wik.example/gpon-self-install"
 }}
→ 201 {"notificationId":"not_01HZ_WANJIKU_SELFINSTALL"}
```

Wanjiku gets the SMS + email. BPMN parks at customer-ont-online message catch, with 14-day boundary timer.

Day 3 — 18:00: Wanjiku picks up the new ONT at Wananchi Shop Westgate. Self-installs at home. Plugs in. OLT auto-discovers the new ONT serial 48575443BEEF5678 on F1/S2/P3. The Wananchi malogs portal logs it. NMS auto-bind subsystem detects + binds (because the serial was pre-registered for Wanjiku's account at instance creation). NMS emits a webhook to Sophix:

```
POST /webhook/nms/ont-online
{"ontSerial":"48575443BEEF5678","oltFsp":{"frame":1,"slot":2,"port":3},
 "boundAccountId":"acct_01HZ_WANJIKU","onlineAt":"2026-05-25T17:58:00+03:00"}
```

Sophix maps the webhook to the awaiting BPMN process instance (correlation: target ONT serial → swap_request_id). Message catch customer-ont-online fires. BPMN advances.

Day 3 — 18:02: Worker nms-remote-bind-stub fires (confirmatory call; auto-discovery already bound):

```
POST /nms-adapter/v1/ont/bind
{"ontSerial":"48575443BEEF5678","oltFsp":{"frame":1,"slot":2,"port":3},"olt":"nrb-olt-01"}
→ 200 {"bindOk":true,"adapterResponseCode":"OK","alreadyBoundViaAutoDiscovery":true}
```

Worker instance-state-transition advances target instance:

```
POST /api/equipment-instances/inst_01HZ_ONT_WANJIKU_NEW/lifecycle-events
{"toState":"IN_FIELD_ACTIVE","toAccountId":"acct_01HZ_WANJIKU",
 "reasonCode":"REMOTE_BIND_OK","contractorId":"wik-internal-tech-team",
 "originRefKind":"swap_request","originRefId":"swap_01HZ_WANJIKU_ONT",
 "actorUserId":"system-nms-autobind","expectedEventSequence":3}
→ 201 {"eventId":"evt_01HZ_WANJIKU_BIND","eventSequence":4}
```

State SOURCE_RECOVERED → TARGET_PLACED.

Day 3 — 18:03: Worker homepass-network-path-update fires:

```
PATCH /api/homepasses/hp_01HZ_NRB_KILELESHA_42B/network-path/ports
{"nodePortUpdates":[{"nodeCode":"48575443ABCD1234","newCode":"48575443BEEF5678"}]}
→ 200 {"updated":1,"previousOntSerial":"48575443ABCD1234","newOntSerial":"48575443BEEF5678"}
```

The HomePass terminal node's code is now the new ONT serial. Kafka emits SwapRequestHomePassUpdated. State TARGET_PLACED → PROVISIONING_IN_PROGRESS → COMPLETED.

Day 3 — 18:05: Wanjiku gets SMS: "Your new ONT is online. Your internet is back. Please drop your old ONT at any Wananchi shop in the next 30 days. Thank you." Swap-request closes. Kafka emits SwapRequestCompleted.

Day 14 — 11:00: Wanjiku drops off the old ONT at the Westgate shop. Shop agent scans the serial, marks it received via the OSR-01 stock receipt API. The receipt triggers a delayed lifecycle event on the source instance:

```
POST /api/equipment-instances/inst_01HZ_ONT_WANJIKU_OLD/lifecycle-events
{"toState":"RECOVERED_BY_CONTRACTOR","toLocationId":"WIK-VAN-wik-internal-tech-team",
 "reasonCode":"CONTRACTOR_RECOVERED_FROM_FIELD","contractorId":"wik-internal-tech-team",
 "originRefKind":"swap_request","originRefId":"swap_01HZ_WANJIKU_ONT",
 "actorUserId":"shop-agent-westgate-01","expectedEventSequence":6}
→ 201 {"eventId":"evt_01HZ_WANJIKU_PICKUP","eventSequence":7}
```

Worker contractor-stock-route writes the stock movement (van location). Defective unit is now in WIK Internal Tech Team's van (or returned to main as part of next sweep, depending on internal-team policy). Kafka emits SwapRequestSourceRecovered with recoveryChannel=COURIER_PICKUP.

The swap-request itself is already COMPLETED since Day 3; the Day 14 event is a side effect that updates instance state without re-opening the swap-request.

8. Worked end-to-end example — field fallback path

Customer: Otieno, WIK fiber subscriber, elderly, not tech-savvy. ONT fails. NOC posts POST /api/swap-requests/gpon. Eligibility passes; remote unassign succeeds; NOC user task: agent talks to Otieno on the phone, Otieno says "I don't want to install anything, send someone." NOC agent submits customerSelfSwapConfirmed=false.

BPMN enters subflow:

```
UPDATE equipment_swap_request SET swap_kind='GPON_FIELD', flow_payload = jsonb_set(flow_payload, '{fieldFallbackTriggered}', 'true')  
WHERE swap_request_id='swa'
```

History row written. State advances to AWAITING_SLOT. From here, the flow mirrors SWAP-HFC's field-visit pattern (slot reserve → WO-01 → tech visit → tech-confirm-defect → instance transitions → NMS bind). The HomePass update happens at TARGET_PLACED, same as the remote-first happy path.

9. Workshop sources

- **Workshop 2026-04-30 (Provisioning deep dive — GPON-specific)** — ONT identified by serial alone (no MAC handshake at BSS level); OLT auto-discovery via malogs portal; F/S/P recorded but not authorization-bound; static-IP GPON manual via DHCP console (out of v1.0 GPON scope). Drives R-GPON-1, R-GPON-2 and the flow_payload shape.
- **Workshop 2026-04-30 (Customer Management deep dive)** — Equipment-routing gap (40+ stale items). Drives the contractor-stock-route worker invocation at courier exchange / shop receipt.
- **Workshop 2026-04-30 (Provisioning deep dive — Service Termination)** — Equipment dismantle = separate retrieval WO. Applied here in the field-fallback subflow.
- **DD_HomePass-CFG §"Equipment / ONT binding"** — HomePass.network_path terminal node carries the ONT serial as its code. Drives R-GPON-5 (mandatory HomePass update at TARGET_PLACED for GPON swaps) and the homepass-network-path-update worker step.
- **DD_HomePass-CFG REST API §"Network path PATCH"** — PATCH /api/homepasses/{id}/network-path/ports accepts node-port updates including code changes. Used by the worker.
- **BP-08 Equipment Swap (RMA) GAP analysis** — Wananchi anchor: GPON swap = remote-first; G-08-1 workflow + G-08-2 contractor stock traceability + provisioning slot/port preservation (not applicable to GPON authentication, but F/S/P recorded for traceability per UC-PR4 spirit).
- **malogs.wananchi.com gpon_auto_find portal** — OLT auto-discovery feed (workshop image: ONT SN + F/S/P listings). Drives the NMS auto-bind webhook pattern in §7 Day 3.
- **Workshop 2026-04-28 (Billing deep dive)** — Wallet funding gate for contractor WOs. Drives pay-first when GPON swap is chargeable (e.g., customer damage to ONT, out-of-warranty); free swaps (warranty / system fault) skip pay-first.